Software Construction and Development

Course Instructor: *Engr. Muhammad Umer Haroon*

Tentative Course Content

Week	Content
1	Software Construction, Clean Code, Meaningful Names
2	Programming Language Constructs & Clean Code Recommendations
3	Reusability: Functions/Lambdas, Coupling & Cohesion
4	Tech Stack Selection, COTS, Continuous Improvement (Releases/Versioning/Patching)
5	Regular Expressions: Good or Bad
6	File Handling & Data Interchange Formats
7	Exception Handling & Defensive Programming
8	Concurrency Programming
9	Data Acquisition for Software Construction (Web Scraping)
10	Design Patterns I: Creational & Structural
11	Design Patterns II: Behavioral
12	Anti-Patterns & Code Smells (Refactoring toward Clean Code)
13	APIs: Bridging Software Systems & Productivity Automation
14	Secure Coding
15	AI-Augmented Software Construction
16	Presentations / Project Evaluation